9h. Mutations and Pre-Allocations

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INTRODUCTION

Certain operations executed with for-loops may trigger the creation of new vectors during each iteration, leading to repeated memory allocation. In many cases, this dynamic allocation is unnecessary. Particularly, when the vectors only store temporary intermediate results that are discarded after the current iteration.

A similar issue arises when the same operations must be executed repeatedly with varying inputs, as in simulations or Monte Carlo experiments. If only the final output is required, generating new intermediate vectors for every run results in avoidable allocations and wasted resources.

In these situations, performance can be improved through **pre-allocation**. This technique consists of initializing a vector before execution begins, then reusing it throughout each computation. By allocating memory upfront and updating the vector in place, the program avoids the repeated overhead of memory allocation.

Remarkably, the technique isn't exclusive to Julia. Rather, it represents an optimization strategy applicable across programming languages. Its effectiveness ultimately stems from **favoring mutations over the creation of new objects**, thus minimizing allocations on the heap.

Our presentation will include a review of methods for initializing vectors. This is a prerequisite for implementing a pre-allocation strategy. We then present some scenarios where pre-allocation proves advantageous, with special emphasis on its advantages within nested for-loops.

THE ADVANTAGES OF MUTATING VECTORS

The benefits of pre-allocation stem from the broader advantages of in-place operations. This strategy encourages reusing existing vectors rather than creating new objects. Because this principle is valuable beyond pre-allocation, we begin by illustrating the performance gains that come from reusing the same memory address.

To make this concrete, consider the operation of sorting a vector `x`. In Julia, many built-in functions have an in-place function counterpart. They're easily identified by inclusion of the symbol `!` at the end of the function name. For example, `sort(x)` allocates a new sorted vector, while `sort!(x)` rearranges the elements of `x` in place. When preserving the original ordering isn't necessary, creating

a separate copy provides no benefit. In those cases, the mutating version `sort!(x)` is preferable, as it avoids unnecessary allocations by reusing the memory associated with `x`. The gains from this approach are presented below.

SORT

```
x = rand(1_000_000)
```

```
julia> @btime sort($x)
  43.514 ms (9 allocations: 15.267 MiB)
```

SORT!

```
x = rand(1_000_000)
```

```
julia> @btime sort!($x)
  1.750 ms (0 allocations: 0 bytes)
```

As we can see, the differences in runtime are substantial. If the same operation were executed repeatedly inside a for-loop, the performance gaps would become even more pronounced.

Pre-allocation builds on the same underlying idea, but applies to situations where a vector like `x` doesn't yet exist. Considering this, we begin with a review of how to initialize vectors efficiently.

INITIALIZING VECTORS

> **Remark**
>
> The review of methods for vector initialization will be relatively brief and centered on performance considerations. For a more detailed coverage, see the section about vector initialization.

Vector initialization refers to the process of creating a vector that subsequently will be filled with values. The process typically involves two steps: reserving space in memory and populating that space with some initial values. An efficient approach to initializing a vector involves performing only the first step, keeping whatever content is in the memory address. These values held in memory are referred to as `undef` (undefined). Although Julia will display them as numerical values, they're essentially arbitrary and meaningless.

There are two methods for initializing a vector with `undef` values. The first one is through a constructor, requiring the specification of the length and element types. The second one is based on the function `similar`, which creates a vector with the same type and dimension as another existing vector. This approach is particularly useful when the output sought matches the structure of an input.

Below, we compare the performance of several approaches to initializing vectors. In particular, we establish that working with `undef` values is faster than populating vectors with specific values. To starkly show the differences in execution time, we repeat the process of vector creation 100,000 times.

UNDEF

```
x           = collect(1:100)
repetitions = 100_000                       # repetitions in a for-loop

function foo(x, repetitions)
    for _ in 1:repetitions
        Vector{Int64}(undef, length(x))
    end
end
```

```
julia> @btime foo($x, $repetitions)
  3.455 ms (100000 allocations: 85.449 MiB)
```

SIMILAR (UNDEF)

```
x           = collect(1:100)
repetitions = 100_000                       # repetitions in a for-loop

function foo(x, repetitions)
    for _ in 1:repetitions
        similar(x)
    end
end
```

```
julia> @btime foo($x, $repetitions)
  2.977 ms (100000 allocations: 85.449 MiB)
```

ZEROS

```
x           = collect(1:100)
repetitions = 100_000                       # repetitions in a for-loop

function foo(x, repetitions)
    for _ in 1:repetitions
        zeros(Int64, length(x))
    end
end
```

```
julia> @btime foo($x, $repetitions)
  10.797 ms (100000 allocations: 85.449 MiB)
```

ONES

```
x           = collect(1:100)
repetitions = 100_000                       # repetitions in a for-loop

function foo(x, repetitions)
    for _ in 1:repetitions
        ones(Int64, length(x))
    end
end
```

```
julia> @btime foo($x, $repetitions)
  11.003 ms (100000 allocations: 85.449 MiB)
```

FILL

```
x           = collect(1:100)
repetitions = 100_000                       # repetitions in a for-loop

function foo(x, repetitions)
    for _ in 1:repetitions
        fill(2, length(x))                  # vector filled with integer 2
    end
end
```

```
julia> @btime foo($x, $repetitions)
  10.170 ms (100000 allocations: 85.449 MiB)
```

> **Remark**
>
> Recall that `_` isn't a keyword, but rather a convention widely adopted by programmers to denote **dummy variables**. These are variables included solely to meet syntactical requirements, but are never actually used or referenced within the code. In our example, the inclusion of `_` is because for-loops must always include a variable to iterate over. While any other symbol could be used, `_` signals programmers that its value can be safely ignored.

INITIALIZING VECTORS IN FUNCTIONS

We can initialize a vector by passing it to the function as a keyword argument. This strategy even enables the use of `similar(x)`, where `x` is a previous function argument. Considering this, the following two implementations are equivalent.

IMPLEMENTATION 1

```
function foo(x)
   output = similar(x)
   # <some calculations using 'output'>
end
```

IMPLEMENTATION 2

```
function foo(x; output = similar(x))

   # <some calculations using 'output'>
end
```

When multiple variables of the same type need to be initialized, array comprehensions offer a concise solution. A more efficient alternative is based on the so-called **generators**, which are the lazy counterpart of array comprehensions. This reduces memory allocations by initializing and assigning the vectors element-wise. The behavior differs from array comprehensions, which first create a vector that holds all the initialized vectors.

Below we illustrate these strategies. Although the example is based on `similar(x)`, note that the same principle applies a constructor like `Vector{Float64}(undef, length(x))`.

ARRAY COMPREHENSION

```
x = [1,2,3]

function foo(x)
    a,b,c = [similar(x) for _ in 1:3]
    # <some calculations using a,b,c>
end
```

```
julia> @btime foo($x)
  101.973 ns (8 allocations: 320 bytes)
```

GENERATOR

```
x = [1,2,3]

function foo(x)
    a,b,c = (similar(x) for _ in 1:3)
    # <some calculations using a,b,c>
end
```

```
julia> @btime foo($x)
  40.751 ns (3 allocations: 144 bytes)
```

DESCRIBING THE PRE-ALLOCATION TECHNIQUE

To describe how pre-allocation works, we'll consider a scenario where the strategy proves to be advantageous: **nested for-loops**. In these constructions, the output of a for-loop serves as an intermediate input for another for-loop. To illustrate them, we'll provide examples with explicit for-loops, but also with operations computed internally via for-loops (e.g., broadcasting).

Let's first describe the inner for-loop. Suppose we're assessing a worker's performance over a 30-day period, with daily scores recorded on a scale from 0 to 1. Defining successful performance as any score above 0.5, the following code snippets generate vectors indicating the days on which the goal was achieved.

BROADCASTING

```
nr_days             = 30
score               = rand(nr_days)

performance(score) = score .> 0.5
```

```
julia> @btime performance($score)
  49.875 ns (3 allocations: 96 bytes)
```

FOR-LOOP (EQUIVALENT)

```
nr_days             = 30
score               = rand(nr_days)

function performance(score)
    target = similar(score)

    for i in eachindex(score)
        target[i] = score[i] > 0.5
    end

    return target
end
```

```
julia> @btime performance($score)
  45.300 ns (2 allocations: 304 bytes)
```

FOR-LOOP (EQUIVALENT)

```
nr_days             = 30
score               = rand(nr_days)

function performance(score; target=similar(score))


    for i in eachindex(score)
        target[i] = score[i] > 0.5
    end

    return target
end
```

```
julia> @btime performance($score)
  44.777 ns (2 allocations: 304 bytes)
```

Consider now that the output of this operation (i.e., the vector of days in which the target was met) represents an intermediate step in another for-loop. For instance, suppose we have performance scores for multiple workers and wish to summarize each worker's data using a summary statistic. Specifically, we'll use the ratio of the standard deviation to the mean, which expresses variability in units of the average.

In practice, calculating this statistic requires both the mean and the standard deviation. Below, we show that computing each statistic via reductions is inefficient. The reason is that, while reductions avoid memory allocations, they require computing which days the target was met twice. Instead, it's more efficient to compute and store this vector beforehand, and then reuse it as the input for both the mean and the standard deviation.

The result highlights a more general principle: when the same intermediate input is required for multiple outputs, the cost of allocating memory for the intermediate vector tends to be lower than its repeated computation. In the example, the inefficient approach is implemented via lazy broadcasting, which internally relies on reduction techniques. The other versions employ explicit or implicit for-loops, storing the intermediate input in a vector.

REDUCTIONS (INEFFICIENT)

```
nr_days             = 30
scores              = [rand(nr_days), rand(nr_days), rand(nr_days)]  # 3 workers



function repeated_call(scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)

      stats[col] = std(@~ scores[col] .> 0.5) / mean(@~ scores[col] .> 0.5)
    end

    return stats
end
```

```
julia> @btime repeated_call($scores)
  685.195 ns (2 allocations: 80 bytes)
```

IMPLICIT NESTED FOR-LOOP

```
nr_days            = 30
scores             = [rand(nr_days), rand(nr_days), rand(nr_days)]   # 3 workers

performance(score) = score .> 0.5

function repeated_call(scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
      target     = performance(scores[col])
      stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call($scores)
  495.696 ns (11 allocations: 368 bytes)
```

EXPLICIT NESTED FOR-LOOP

```
nr_days            = 30
scores             = [rand(nr_days), rand(nr_days), rand(nr_days)]   # 3 workers

function performance(score)
    target = similar(score)

    for i in eachindex(score)
        target[i] = score[i] > 0.5
    end

    return target
end

function repeated_call(scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
      target     = performance(scores[col])
      stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call($scores)
  362.846 ns (8 allocations: 992 bytes)
```

EXPLICIT NESTED FOR-LOOP

```
nr_days            = 30
scores             = [rand(nr_days), rand(nr_days), rand(nr_days)]    # 3 workers

function performance(score; target = similar(score))


    for i in eachindex(score)
        target[i] = score[i] > 0.5
    end

    return target
end

function repeated_call(scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
      target     = performance(scores[col])
      stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call($scores)
  376.145 ns (8 allocations: 992 bytes)
```

Since storing the intermediate vector is more performant, an efficient approach will necessarily involve memory allocations. Yet, there's still room for improvement, as the methods described above allocate more than required. In particular, during each iteration, a new vector is created for every worker to store the days the target was met. Such allocations are unnecessary, since this intermediate vector doesn't need to be preserved for future use. Indeed, the examples store this vector in a local variable, implying that it's dropped once the iteration concludes.

Such cases are strong candidates for pre-allocating intermediate results. By defining a single vector `target`, we can reuse the same object across iterations to compute the days when the target was met for each worker. This strategy incurs the cost of creating the vector only once, after which it's reused for every worker.

The implementation requires initializing `target` before the outer for-loop begins. Then, during each iteration, we mutate `target` using an in-place function, which we'll call `performance!`. This function can be defined to update contents either with a standard for-loop or by applying the broadcasting operator `.=`.

NO PRE-ALLOCATION

```
nr_days = 30
scores  = [rand(nr_days), rand(nr_days), rand(nr_days)]


performance(score) = score .> 0.5

function repeated_call(scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
        target     = performance(scores[col])
        stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call($scores)
  524.400 ns (11 allocations: 368 bytes)
```

PRE-ALLOCATION (VIA BROADCASTING)

```
nr_days = 30
scores  = [rand(nr_days), rand(nr_days), rand(nr_days)]
target  = similar(scores[1])

performance!(target, score) = (@. target = score > 0.5)

function repeated_call!(target, scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
        performance!(target, scores[col])
        stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call!($target, $scores)
  358.638 ns (2 allocations: 80 bytes)
```

PRE-ALLOCATION (VIA FOR-LOOP)

```
nr_days = 30
scores  = [rand(nr_days), rand(nr_days), rand(nr_days)]
target  = similar(scores[1])

function performance!(target, score)
    for i in eachindex(score)
        target[i] = score[i] > 0.5
    end
end

function repeated_call!(target, scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
        performance!(target, scores[col])
        stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call!($target, $scores)
  339.744 ns (2 allocations: 80 bytes)
```

> ⚠️ **Warning!** - Use of @. to update values
>
> When your goal is to update the values of a vector, recall that `@.` has to be *placed at the beginning* of the statement.
>
> **NEW VARIABLE (NOT AN UPDATE)**
>
> ```
> # the following are equivalent and define a new variable
>    output  = @. 2  * x
>    output  =    2 .* x
> ```
>
> **AN UPDATE**
>
> ```
> # the following are equivalent and update 'output'
> @. output  = 2  * x
>    output .= 2 .* x
> ```

Compared to a for-loop, the method using `.=` provides a simpler syntax. For this reason, when the function `performance!` is simple enough as in our example, it's common to skip its definition and directly insert the operation into the inner for-loop. The approach also enables implementing the update via a built-in in-place function. For instance, `map!` can be used with this purpose.

NO PRE-ALLOCATION

```
nr_days = 30
scores  = [rand(nr_days), rand(nr_days), rand(nr_days)]


function repeated_call(scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
        target     = @. score > 0.5
        stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call($scores)
  822.471 ns (23 allocations: 608 bytes)
```

PRE-ALLOCATION (VIA BROADCASTING)

```
nr_days = 30
scores  = [rand(nr_days), rand(nr_days), rand(nr_days)]
target  = similar(scores[1])

function repeated_call!(target, scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
        @. target  = scores[col] > 0.5
        stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call!($target, $scores)
  336.372 ns (2 allocations: 80 bytes)
```

PRE-ALLOCATION (VIA MAP!)

```
nr_days = 30
scores  = [rand(nr_days), rand(nr_days), rand(nr_days)]
target  = similar(scores[1])

function repeated_call!(target, scores)
    stats = Vector{Float64}(undef, length(scores))

    for col in eachindex(scores)
        map!(a -> a > 0.5, target, scores[col])
        stats[col] = std(target) / mean(target)
    end

    return stats
end
```

```
julia> @btime repeated_call!($target, $scores)
  329.814 ns (2 allocations: 80 bytes)
```